

HDOC Day-43

Menu-Driven String Operations in C

Problem Statement

Write a menu-driven C program using separate functions to (1) find the longest word in a sentence and (2) determine whether one string is a rotation of another. Include an Exit option.

Algorithm

1. Display the menu repeatedly until Choice 3 (Exit).
2. For Choice 1, read a sentence using `fgets()`. Scan characters and identify consecutive letters/digits as words. Track the first longest word and its length; display the result or a no-word message.
3. For Choice 2, read two strings. If their lengths differ, report that they are not rotations. Otherwise, concatenate the first string with itself and use `strstr()` to search for the second string.
4. For invalid choices, display an error and show the menu again.

Test Case Table

TC	Choice	Input	Expected result
1	1	C programming is interesting	Longest word: programming; Length: 13
2	1	I love computers	Longest word: computers; Length: 10
3	1	a bb ccc	Longest word: ccc; Length: 3
4	1	!!! ...	No word found in the sentence.
5	2	ABCD / CDAB	Strings are rotations.
6	2	ABCD / ACBD	Strings are not rotations.
7	2	hello / hello	Strings are rotations (zero shift)
8	2	abc / ab	Strings are not rotations (different lengths)
9	9	Invalid menu choice	Invalid choice. Enter 1 to 3.
10	3	Exit	Exiting program.

C Program (hdoc43.c)

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

#define MAX 500

void readLine(char text[], size_t size);
void findLongestWord(void);
void checkRotation(void);

int main(void)
{
    char line[64];
    int choice;
    do {
        printf("\n--- STRING OPERATIONS MENU ---\n");
```

```

    printf("1. Find the longest word in a sentence\n");
    printf("2. Check whether one string is a rotation of another\n");
    printf("3. Exit\n");
    printf("Enter your choice: ");
    readLine(line, sizeof line);
    if (sscanf(line, "%d", &choice) != 1) {
        printf("Invalid input. Enter a number from 1 to 3.\n");
        choice = 0;
    }
    switch (choice) {
        case 1: findLongestWord(); break;
        case 2: checkRotation(); break;
        case 3: printf("Exiting program.\n"); break;
        default: if (choice != 0) printf("Invalid choice. Enter 1 to 3.\n");
    }
} while (choice != 3);
return 0;
}

```

```

void readLine(char text[], size_t size)
{
    size_t len;
    int ch;
    if (fgets(text, (int)size, stdin) == NULL) {
        text[0] = '\0';
        return;
    }
    len = strlen(text);
    if (len > 0 && text[len - 1] == '\n') {
        text[len - 1] = '\0';
    } else if (len == size - 1) {
        while ((ch = getchar()) != '\n' && ch != EOF) { }
    }
}

```

```

void findLongestWord(void)
{
    char sentence[MAX], longest[MAX] = "";
    size_t i = 0, best = 0;
    printf("Enter a sentence: ");
    readLine(sentence, sizeof sentence);
    while (sentence[i] != '\0') {
        size_t start, length;
        while (sentence[i] != '\0' && !isalnum((unsigned char)sentence[i])) i++;
        start = i;
        while (sentence[i] != '\0' && isalnum((unsigned char)sentence[i])) i++;
        length = i - start;
        if (length > best) {
            memcpy(longest, sentence + start, length);
            longest[length] = '\0';
            best = length;
        }
    }
    if (best == 0) printf("No word found in the sentence.\n");
    else printf("Longest word: %s\nLength: %zu\n", longest, best);
}

```

```

void checkRotation(void)
{
    char first[MAX], second[MAX], doubled[2 * MAX];
    size_t n;
    printf("Enter first string: ");
    readLine(first, sizeof first);
}

```

```

printf("Enter second string: ");
readLine(second, sizeof second);
n = strlen(first);
if (n != strlen(second)) {
    printf("The strings are not rotations of each other.\n");
    return;
}
strcpy(doubled, first);
strcat(doubled, first);
if (strstr(doubled, second) != NULL)
    printf("The strings are rotations of each other.\n");
else
    printf("The strings are not rotations of each other.\n");
}

```

Compilation and Execution

```

gcc hdoc43.c -o hdoc43
./hdoc43

```

Compilation and Execution

```

(kali㉿kali)-[~/Desktop]
$ vi hdoc43.c

(kali㉿kali)-[~/Desktop]
$ gcc hdoc43.c -o hdoc43

(kali㉿kali)-[~/Desktop]
$ ./hdoc43

```

Test Case 1 –

```

3. EXIT
Enter your choice: 1
Enter a sentence: C programming is interesting
Longest word: programming
Length: 11

```

Test Case 2 -

```

Enter your choice: 1
Enter a sentence: I love computers
Longest word: computers
Length: 9

```

Test Case 3 -

```
Enter your choice: 1
Enter a sentence: a bb ccc
Longest word: ccc
Length: 3
```

Test Case 4 -

```
Enter your choice: 1
Enter a sentence: !!! ...
No word found in the sentence.
```

Test Case 5 -

```
Enter your choice: 2
Enter first string: ABCD
Enter second string: CDAB
The strings are rotations of each other.
```

Test Case 6 -

```
Enter your choice: 2
Enter first string: ABCD
Enter second string: ACBD
The strings are not rotations of each other.
```

Test Case 7 -

```
Enter your choice: 2
Enter first string: hello
Enter second string: hello
The strings are rotations of each other.
```

Test Case 8 -

```
Enter your choice: 2
Enter first string: abc
Enter second string: ab
The strings are not rotations of each other.
```

Test Case 9 -

```
Enter your choice: 9
Invalid choice. Enter 1 to 3.
```

Test Case 10 -

3. EXIT

Enter your choice: 3
Exiting program.